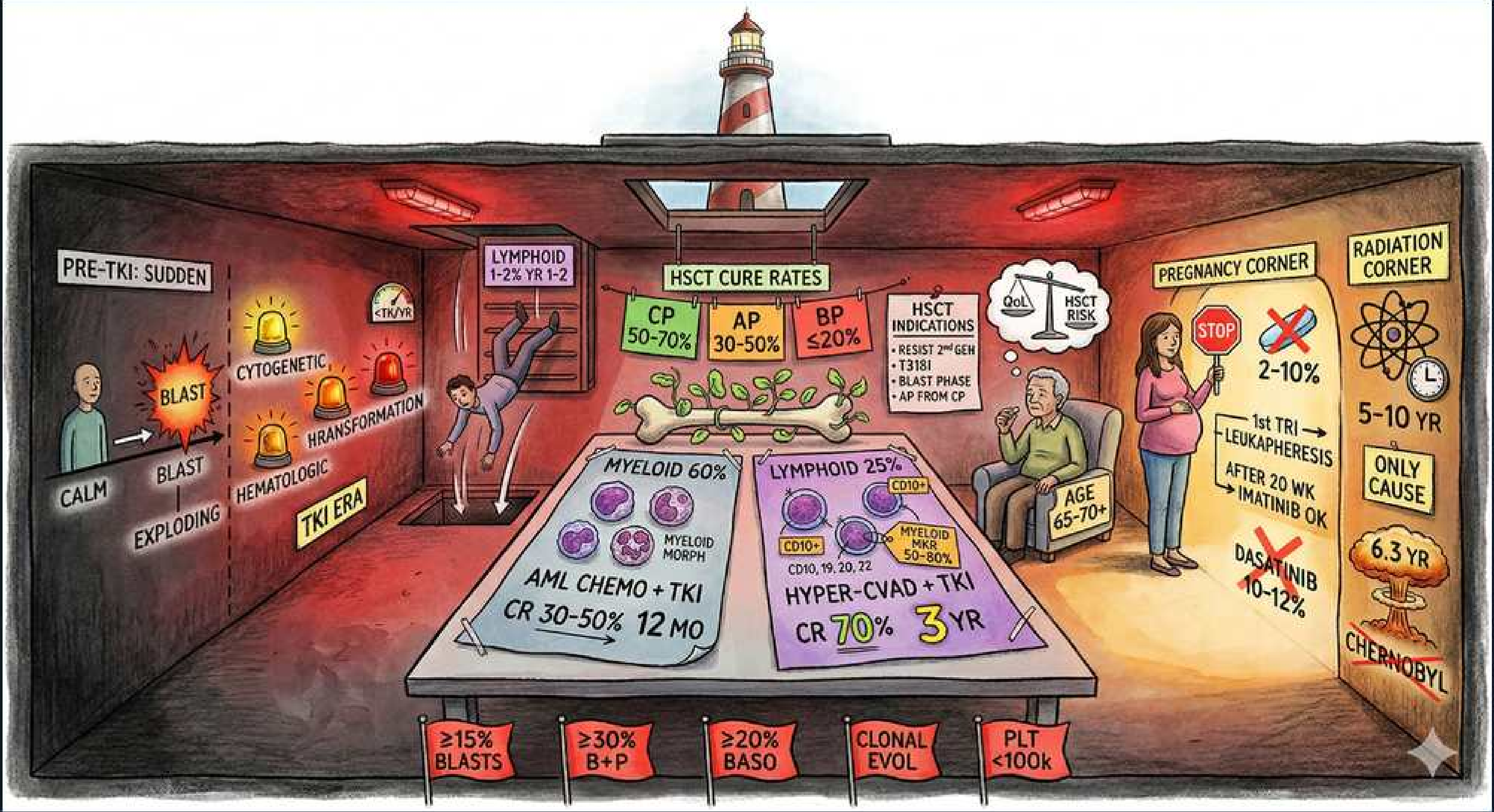
CML — Blast Crisis, Special Situations & Transplant



1. Pre-TKI sudden blast

WHAT YOU SEE

Left wall: PRE-TKI SUDDEN, CALM then BLAST EXPLODING, cytogenetic/hematologic/transformation alarms.

WHAT IT ENCODES

Historically, chronic-phase CML could suddenly transform to blast crisis. Warning signs are cytogenetic (new ACAs), hematologic (rising counts/basophilia), and clinical transformation. TKIs greatly reduced this abrupt progression.

BOARD TRAP

In the TKI era sudden blast crisis is uncommon and usually flagged early by rising PCR or new ACAs — surveillance catches it before overt crisis.

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2. Myeloid blast crisis 60%

WHAT YOU SEE

Center table left: MYELOID 60%, AML CHEMO + TKI, CR 30–50%, 12 MO.

WHAT IT ENCODES

~60% of blast crises are myeloid. Treat with AML-type induction chemotherapy PLUS a TKI; CR rates are only ~30–50% and median survival roughly 12 months. Allogeneic transplant is the goal if a response is achieved.

BOARD TRAP

Myeloid blast crisis responds worse than lymphoid — its ~30–50% CR and ~12-month survival are inferior to lymphoid blast crisis outcomes.

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3. Lymphoid blast crisis 25%

WHAT YOU SEE

Center table right: LYMPHOID 25%, CD10/19/20/22, HYPER-CVAD + TKI, CR 70%, 3 YR.

WHAT IT ENCODES

~25% are lymphoid (B-lineage, CD10/19/22+), resembling Ph+ ALL. Treat with ALL-type therapy (e.g., hyper-CVAD) plus a TKI; CR ~70% with better survival (~few years). Dasatinib/ponatinib penetrate CNS.

BOARD TRAP

Lymphoid blast crisis (Ph+ ALL-like) carries the BETTER prognosis of the two — opposite of intuition; pick ALL-directed chemo, not AML chemo.

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4. Accelerated/phase criteria

WHAT YOU SEE

Bottom red tags: ≥15% BLASTS, ≥30% B+P, ≥20% BASO, CLONAL EVOL, PLT <100k.

WHAT IT ENCODES

WHO/ELN accelerated-phase criteria: 15–29% blasts (or ≥30% blasts+promyelocytes), ≥20% basophils, persistent thrombocytopenia <100k unrelated to therapy, or clonal evolution (new ACAs).

BOARD TRAP

Therapy-RELATED thrombocytopenia does not count — accelerated-phase thrombocytopenia must be disease-related; and ≥20% basophils (not blasts) is its own criterion.

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5. HSCT cure rates by phase

WHAT YOU SEE

HSCT CURE RATES box: CP 50–70%, AP 30–50%, BP ≤20%.

WHAT IT ENCODES

Allogeneic stem cell transplant cure rates fall by phase: chronic 50–70%, accelerated 30–50%, blast ≤20%. Indications: ≥2 TKI failure, T315I not controlled, accelerated/blast phase. Best results transplanting in chronic phase.

BOARD TRAP

Transplant is no longer first-line in chronic phase (TKIs are) but is reserved for TKI failure/T315I/advanced phase — and works best if you can return the patient to chronic phase first.

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6. Pregnancy — avoid TKIs

WHAT YOU SEE

PREGNANCY CORNER: STOP sign over pills, 2–10%, 1st TRI leukapheresis, after 20 wk imatinib OK, dasatinib 10–12%.

WHAT IT ENCODES

TKIs are teratogenic (especially first trimester). For symptomatic disease in pregnancy use leukapheresis +/- interferon-alpha (safe); IFN is the preferred cytoreductive agent. Imatinib may be cautiously considered after organogenesis (~>20 wk) if needed.

BOARD TRAP

Interferon-alpha — not a TKI — is the safe cytoreductive drug in pregnancy; stop TKIs preconception/first trimester. Dasatinib has notable transplacental/fetal risk.

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7. Radiation — only proven cause

WHAT YOU SEE

RADIATION CORNER: ONLY CAUSE, 5–10 yr latency, 6.3 rg, CHERNOBYL.

WHAT IT ENCODES

Ionizing radiation is the only well-established environmental risk factor for CML, with a latency of roughly 5–10 years (atomic-bomb and Chernobyl data).

BOARD TRAP

Benzene and prior chemotherapy cause AML/MDS, not classically CML — radiation is the CML-specific exposure tested.

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8. Age 65–70 typical

WHAT YOU SEE

Elderly figure, AGE 65–70.

WHAT IT ENCODES

CML is a disease of older adults (median ~55–65, many 65–70). Older age and comorbidities steer drug choice — favoring tolerable agents and away from vasculotoxic ponatinib/nilotinib.

BOARD TRAP

In an older comorbid patient, drug selection is driven by toxicity profile (avoid nilotinib/ponatinib CV risk), not just potency.
